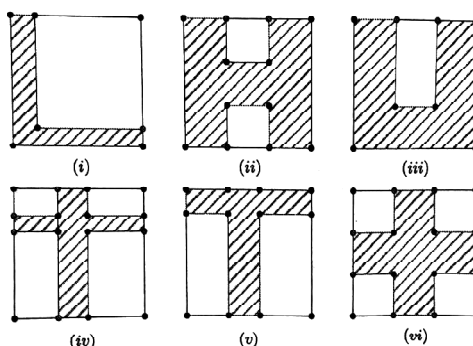
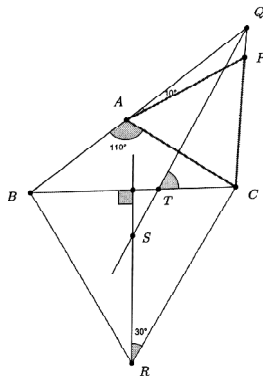


- There are 4 girls and 2 boys of different ages. The eldest is 10 years old while the youngest is 4 years old. The older of the boys is 4 years older than the youngest of the girls. The oldest of the girls is 4 years older than the youngest of the boys. What is the age of the oldest of the boys ?
- In the equation  $A + M + T + I = 10$ .  $A, M, T, I$  are all different natural numbers.  $A$  is the least. Calculate the maximum and minimum value of  $A \cdot M \cdot T \cdot I + A \cdot M \cdot T + A \cdot T \cdot I + M \cdot I \cdot T + M \cdot T \cdot I$  (where  $\cdot$  means multiplication. i.e.,  $A \cdot T \cdot I = A \times T \times I$ ).
- The six squares below are identical. The dimensions of the shaded portions are not known. The perimeter of which shaded areas are equal to the perimeter of the square ? Show the calculations clearly and if the perimeter of any shaded area is different from that of the square, state whether it is more or less than the perimeter of the square.



- $ABD$  is a triangle in which  $\angle A = 110^\circ$ ,  $AB = AC$ .  $APC$  and  $BRC$  are equilateral triangles drawn respectively on  $AC$  and  $BC$  outside the triangle  $ABC$ .  $BA$  is produced and meets  $CP$  produced at  $Q$ . The bisectors of  $\angle Q$  and  $\angle R$  cut at  $S$ . Calculate  $\angle QSR$ . What can you say about the figure  $SRCQ$  ?



- Two numbers are respectively 20% and 50% more than a third number. What percentage is the first of the second ?
  - Three vessels of sizes 3 litres, 4 litres and 5 liters contain mixture of water and milk in the ratio 2 : 3, 3 : 7 and 4 : 11 respectively. The contents of all the three vessels are poured into a single vessel. What is the ratio of water to milk in the resultant mixture ?
- It is a well-known fact that Mahatma Gandhi was the man responsible for getting us the freedom. We got independence in 1947. Mahatma was born in 1869. Find the smallest numbers by which
  - 1869 should be multiplied to get a product which ends in 1947.
  - 1947 should be multiplied to get a product which ends in 1869.
 (The method you use to obtain the required numbers should also be given).

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